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(54) **METHOD AND SYSTEM TO ANALYZE SPACES FOR CLEANING BASED ON THE PRESENCE OF AN INDIVIDUAL**

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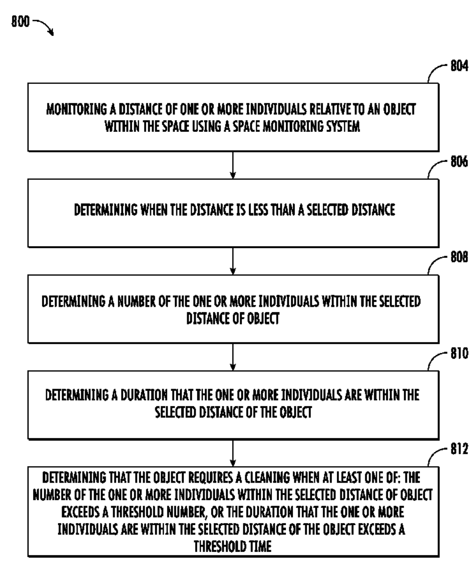
(57) **ABSTRACT**

A method for monitoring and controlling a cleanliness of a space including: monitoring a distance of one or more individuals relative to an object within the space using a space monitoring system; determining when the distance is less than a selected distance; determining a number of the one or more individuals within the selected distance of object; determining a duration that the one or more individuals are within the selected distance of the object; and determining that the object requires a cleaning when at least one of: the number of the one or more individuals within the selected distance of object exceeds a threshold number, or the duration that the one or more individuals are within the selected distance of the object exceeds a threshold time.

17 Claims, 2 Drawing Sheets



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G08B 21/18	(2006.01)

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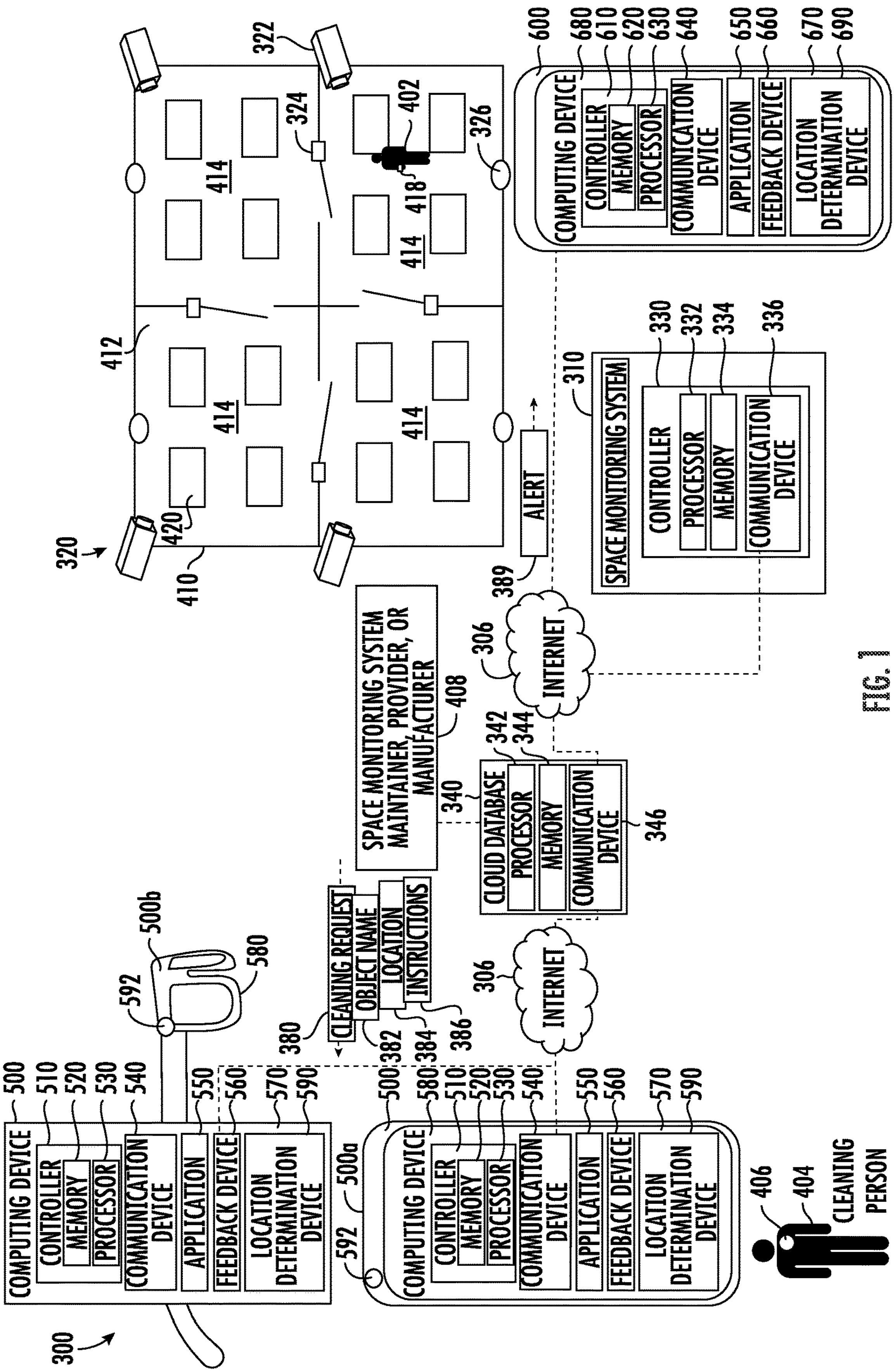
300
320
322
324
326
402
406
404
CLEANING PERSON
408
410
412
414
418
420
ALERT
389
306
INTERNET
310
SPACE MONITORING SYSTEM
CONTROLLER
PROCESSOR
MEMORY
COMMUNICATION DEVICE
330
332
334
336
SPACE MONITORING SYSTEM MAINTAINER, PROVIDER, OR MANUFACTURER
340
342
344
346
CLOUD DATABASE
380
382
384
386
CLEANING REQUEST
OBJECT NAME
LOCATION
INSTRUCTIONS
500
500a
500b
510
520
530
540
550
560
570
580
590
592
COMPUTING DEVICE
APPLICATION
FEEDBACK DEVICE
LOCATION DETERMINATION DEVICE
600
610
620
630
640
650
660
670
680
690

FIG. 1

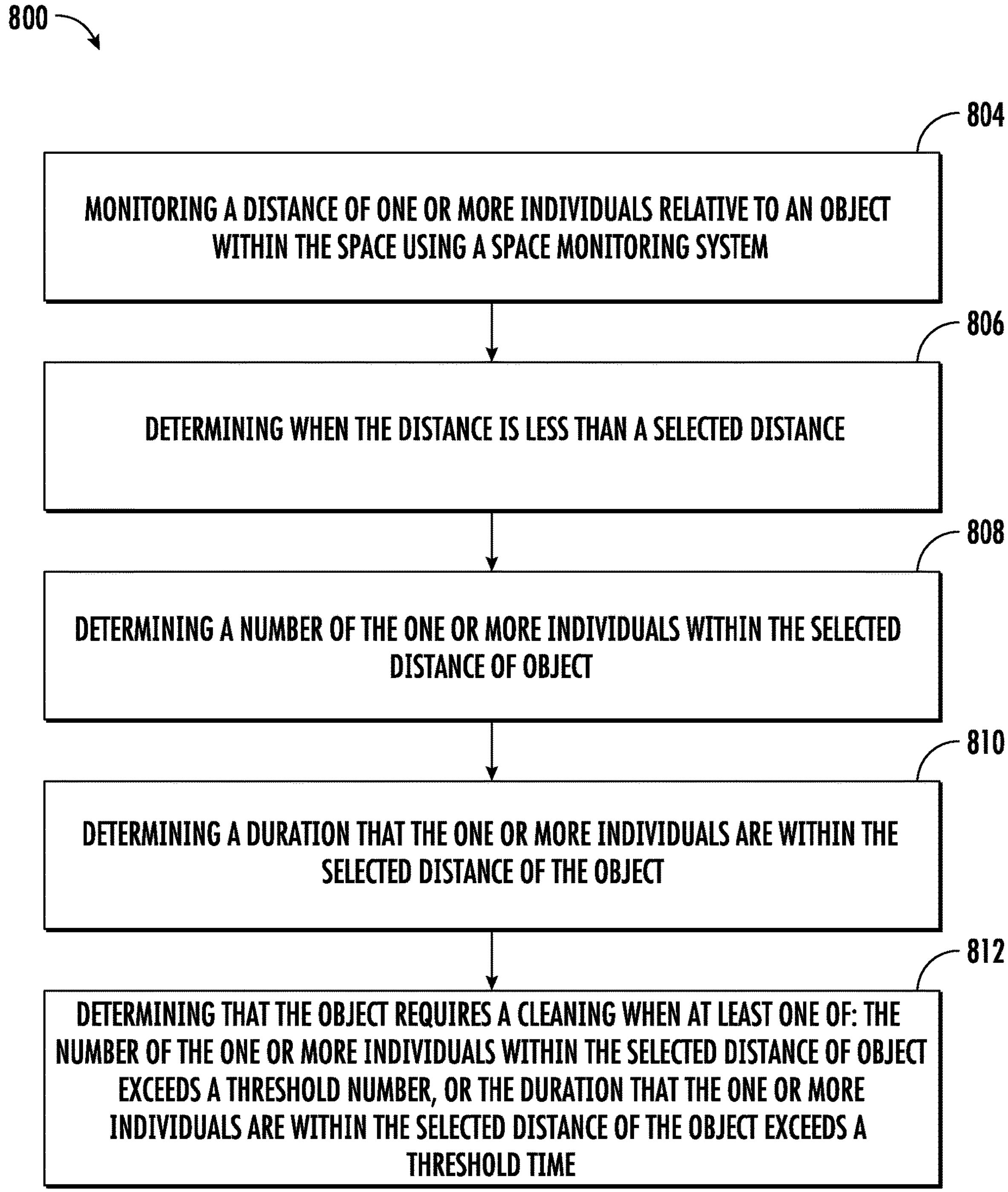
800
804
MONITORING A DISTANCE OF ONE OR MORE INDIVIDUALS RELATIVE TO AN OBJECT WITHIN THE SPACE USING A SPACE MONITORING SYSTEM
806
DETERMINING WHEN THE DISTANCE IS LESS THAN A SELECTED DISTANCE
808
DETERMINING A NUMBER OF THE ONE OR MORE INDIVIDUALS WITHIN THE SELECTED DISTANCE OF OBJECT
810
DETERMINING A DURATION THAT THE ONE OR MORE INDIVIDUALS ARE WITHIN THE SELECTED DISTANCE OF THE OBJECT
812
DETERMINING THAT THE OBJECT REQUIRES A CLEANING WHEN AT LEAST ONE OF: THE NUMBER OF THE ONE OR MORE INDIVIDUALS WITHIN THE SELECTED DISTANCE OF OBJECT EXCEEDS A THRESHOLD NUMBER, OR THE DURATION THAT THE ONE OR MORE INDIVIDUALS ARE WITHIN THE SELECTED DISTANCE OF THE OBJECT EXCEEDS A THRESHOLD TIME

FIG. 2

METHOD AND SYSTEM TO ANALYZE SPACES FOR CLEANING BASED ON THE PRESENCE OF AN INDIVIDUAL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 63/320,376, filed Mar. 16, 2022, all of which are incorporated herein by reference in their entirety.

BACKGROUND

The embodiments herein generally relate to cleaning of objects and more specifically to a method and apparatus for monitoring a presence of individuals in proximity of an object to determine whether cleaning of said object is required.

Cleaning persons often spend countless hours cleaning entire spaces without knowing whether individuals were even present in certain locations of said space. Cleaning of spaces is often based on set schedules regardless of whether the space is being used.

BRIEF SUMMARY

According to one embodiment, a method for monitoring and controlling a cleanliness of a space is provided. The method including: monitoring a distance of one or more individuals relative to an object within the space using a space monitoring system; determining when the distance is less than a selected distance; determining a number of the one or more individuals within the selected distance of object; determining a duration that the one or more individuals are within the selected distance of the object; and determining that the object requires a cleaning when at least one of: the number of the one or more individuals within the selected distance of object exceeds a threshold number, or the duration that the one or more individuals are within the selected distance of the object exceeds a threshold time.

In addition to one or more of the features described above, or as an alternative, further embodiments may include generating a cleaning request.

In addition to one or more of the features described above, or as an alternative, further embodiments may include transmitting the cleaning request to a computing device of a cleaning person.

In addition to one or more of the features described above, or as an alternative, further embodiments may include that the cleaning request includes at least one of an object name of the object, a location of the object, or instructions detailing how to clean the object.

In addition to one or more of the features described above, or as an alternative, further embodiments may include that monitoring the distance of the one or more individuals relative to the object within the space using the space monitoring system further includes: detecting the distance of one or more individuals relative to the object within the space using a camera.

In addition to one or more of the features described above, or as an alternative, further embodiments may include that monitoring the distance of the one or more individuals relative to the object within the space using the space monitoring system further includes: detecting the distance of one or more individuals relative to the object within the space using a door access control device.

In addition to one or more of the features described above, or as an alternative, further embodiments may include that monitoring the distance of the one or more individuals relative to the object within the space using the space monitoring system further includes: detecting the distance of one or more individuals relative to the object within the space using a wireless signal tracking device.

In addition to one or more of the features described above, or as an alternative, further embodiments may include that the wireless signal tracking device is configured to detect an advertisement of a wearable tracking device worn by the one or more individuals or wireless connect to the wearable tracking device worn by the one or more individuals.

In addition to one or more of the features described above, or as an alternative, further embodiments may include providing the cleaning request to a cleaning person through a computing device.

In addition to one or more of the features described above, or as an alternative, further embodiments may include displaying the cleaning request for a cleaning person on a display device of the cleaner computing device.

In addition to one or more of the features described above, or as an alternative, further embodiments may include: capturing an image of the object on an augmented reality capable camera of the cleaner computing device; displaying the image of the object on a display device of the cleaner computing device while using the augmented reality capable camera; and identifying the object that requires the cleaning using augmented reality while using the augmented reality capable camera.

In addition to one or more of the features described above, or as an alternative, further embodiments may include that the cleaner computing device is a smart phone.

In addition to one or more of the features described above, or as an alternative, further embodiments may include that the cleaner computing device is a pair of smart glasses.

In addition to one or more of the features described above, or as an alternative, further embodiments may include monitoring the cleaning of the object performed by a cleaning person using a wearable camera.

In addition to one or more of the features described above, or as an alternative, further embodiments may include providing further cleaning instructions, a compliment, or a cleaning review based on the cleaning of the object performed by the cleaning person.

In addition to one or more of the features described above, or as an alternative, further embodiments may include monitoring a cleaning of the object performed by a cleaning person using an augmented reality capable camera of a pair of smart glasses.

In addition to one or more of the features described above, or as an alternative, further embodiments may include providing further cleaning instructions, a compliment, or a cleaning review based on the cleaning of the object performed by the cleaning person.

In addition to one or more of the features described above, or as an alternative, further embodiments may include transmitting an alert to a computing device for an individual to avoid the object in response to determining that the object requires the cleaning.

According to another embedment, a space cleanliness analysis system for monitoring and controlling a cleanliness of a space is provided. The space cleanliness analysis system including: a processor; and a memory including computer-executable instructions that, when executed by the processor, cause the processor to perform operations, the operations including: monitoring a distance of one or more individuals relative to an object within the space using a space monitoring system; determining when the distance is less than a selected distance; determining a number of the one or more individuals within the selected distance of object; determining a duration that the one or more individuals are within the selected distance of the object; and generating that the object requires a cleaning when at least one of: the number of the one or more individuals within the selected distance of object exceeds a threshold number, or the duration that the one or more individuals are within the selected distance of the object exceeds a threshold time.

According to another embedment, a computer program product tangibly embodied on a non-transitory computer readable medium, the computer program product including instructions that, when executed by a processor, cause the processor to perform operations including: monitoring a distance of one or more individuals relative to an object within the space using a space monitoring system; determining when the distance is less than a selected distance; determining a number of the one or more individuals within the selected distance of object; determining a duration that the one or more individuals are within the selected distance of the object; and generating that the object requires a cleaning when at least one of: the number of the one or more individuals within the selected distance of object exceeds a threshold number, or the duration that the one or more individuals are within the selected distance of the object exceeds a threshold time.

Technical effects of embodiments of the present disclosure include monitoring the relative location of individuals in relation to objects and their duration at said relative location to determine whether the object requires a cleaning.

The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, that the following description and drawings are intended to be illustrative and explanatory in nature and non-limiting.

BRIEF DESCRIPTION

The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

FIG. **1** is a block diagram of an exemplary space cleanliness analysis system, according to an embodiment of the present disclosure; and

FIG. **2** is a flow diagram illustrating an exemplary method for monitoring and controlling a cleanliness of a space, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

Referring to FIG. **1** a schematic diagram of an exemplary space cleanliness analysis system **300** is illustrated, according to an embodiment of the present disclosure. It should be appreciated that, although particular systems are separately defined in the schematic block diagrams, each or any of the systems may be otherwise combined or separated via hardware and/or software.

The space cleanliness analysis system **300**, as illustrated, may include the cloud database **340**, a space monitoring system **310**, and a computer application **550** installed or accessible on a cleaner computing device **500**. The space monitoring system **310** includes one or more space monitoring devices **320** that may be distributed throughout an internal space **412** of a building **410**. It is understood that while the following description associated figures describe an internal space **412** of a building **410**, the embodiment disclosed herein may be applied to any space or area, including, but not limited to, external spaces.

The building **410** may be broken up into one or more rooms **414** and there may be at least one or more space monitoring devices **320** located within each of the rooms **414**. The building **410** may be a home, an apartment, a business, an office building, a hotel, a sports facility, a garage, a room, a shed, a boat, a plane, a bus, elevator car interior, classrooms, libraries, university buildings, or any other structure known to one of skill in the art.

The space monitoring device **320** is configured to monitor the location of an individual **402** within the internal space **412** and/or a time period the individual **402** spends at the location. The space monitoring device **320** may be configured to differentiate one individual **402** from another to determine a number of individuals **402**. The space monitoring device **320** may also be configured to monitor a location **384** of an object **420** within the internal space **412** and/or the location **384** of the object **420** may be saved to a memory **334**. The space monitoring device **320** may also be configured to determine an object name **382** of the object **420** within the internal space **412** and/or the object name **382** of the object **420** may be saved to a memory **334**. The space monitoring device **320** may be able to determine the location **384** of the object **420** and/or the object name **382** of the object **420** using machine learning that is performed by the space monitoring device **320**, the space monitoring system **310**, and/or the cloud database **340**.

The space monitoring device **320** may include a camera **322**, a door access control device **324**, and/or a wireless signal tracking device **326**.

The camera **322** may be configured to capture images of the individuals **402** located within the internal space **412**. The camera **322** may be a video camera, fish-eye camera, or a thermal imaginning camara. In one example, the fish-eye camera may be configured to analyze people density based on heat map and intersection analysis. In embodiment, the camera is configured to analyze people density and capture credentials of an individual **402** to determine who they are or what their role may be in the building **410** (e.g., janitor). The controller **330**, using the camera **322**, is configured to monitor a location of the individual **402** in the room **414** and how long the individual **402** spent at the location in the room **414**. In one embodiment, the camera **322** may be configured to capture images of the objects **420** in order to determine a location of the individual **402** relative to the object **420**. In another embodiment, the location of the objects **420** may be saved in the space monitoring system **310** and the location of the objects **420** does not need to be determined by the space monitoring device **320**.

The door access control device **324** may be a badge reader operably connected to a door lock or turnstile lock. The controller **330**, using the door access control device **324**, is configured to monitor when the individual **402** enters the room **414**, when the individual **402** leaves the room **414**, and how long the individual **402** spent in the room **414**.

The wireless signal tracking device **326** may be configured to track a location of the individual **402** throughout the internal space **412** using wireless triangulation, received signal strength indication (RSSI) and/or Time-of Flight wireless signal monitoring. The wireless signal tracking device **326** may wireless connect to or detect wireless advertisement from a wearable tracking device **418** that is worn by the individual **402**. The wearable tracking device **418** may be a wearable user device, a wrist band, a mobile application, a smart watch, or any other similar device known to one of skill in the art. In order to detect or communicate with the wearable tracking device **418**, the wireless signal tracking device **326** may utilize Bluetooth, Bluetooth Low Energy, Ultra-wideband, Zigbee, Near Field Communication (NFC), or any other similar communication method known to one of skill in the art. Likewise, the wearable tracking device **418** may utilize Bluetooth, Bluetooth Low Energy, Ultra-wideband, Zigbee, Near Field Communication (NFC), or any other similar communication method known to one of skill in the art. The controller **320**, using the wireless signal tracking device **326**, is configured to monitor a location of the individual **402** in the room **414** and how long the individual **402** spent at the location in the room **414**.

The space monitoring system **310** includes a controller **330**. The controller **330** for the space monitoring system **310** may be an internet of things (IoT) connected device.

The controller **330** is configured to communicate with the computer application **550** and the cloud database **340**. The controller **330** may be configured to communicate with the computer application **550** through the cloud database **340**. The controller **330** may be an electronic controller including a processor **332** and an associated memory **334** comprising computer-executable instructions (i.e., computer program product) that, when executed by the processor **332**, cause the processor **332** to perform various operations. The processor **332** may be, but is not limited to, a single-processor or multi-processor system of any of a wide array of possible architectures, including field programmable gate array (FPGA), central processing unit (CPU), application specific integrated circuits (ASIC), digital signal processor (DSP) or graphics processing unit (GPU) hardware arranged homogenously or heterogeneously. The memory **334** may be but is not limited to a random access memory (RAM), read only memory (ROM), or other electronic, optical, magnetic or any other computer readable medium.

The controller **330** also includes a communication device **336**. The communication device **336** may be capable of wireless communication including but not limited to Wi-Fi, Bluetooth, Zigbee, Sub-GHz RF Channel, cellular, satellite, or any other wireless signal known to one of skill in the art. The communication device **336** may be configured to communicate with the cloud database **340** through the internet **306** using the communication device **336**. The communication device **336** may be connected to the internet **306** through a Wi-Fi router or home automation system(not shown). Alternatively, or additionally, the communication device **336** may be configured to communicate directly with the cloud database **340**.

The cloud database **340** may belong to and/or be managed by a space cleanliness monitoring system maintainer, provider, or manufacturer **408**, such as, for example a manufacturer of the space monitoring system **310**, a provider of the space monitoring system **310**, a third-party service provider, or any service provider that may maintain the space monitoring system **310**. The space cleanliness monitoring system maintainer, provider, or manufacturer **408** may be a person, an organization, a group, a partnership, a company, or a corporation.

In an alternate embodiment, the cloud database **340** may be distributed amongst multiple cloud databases rather than the single cloud database **340** that is illustrated in FIG. **1**.

The cloud database **340** may be a remote computer server that includes a processor **342** and an associated memory **344** comprising computer-executable instructions (i.e., computer program product) that, when executed by the processor **342**, cause the processor **342** to perform various operations. The processor **342** may be, but is not limited to, a single-processor or multi-processor system of any of a wide array of possible architectures, including field programmable gate array (FPGA), central processing unit (CPU), application specific integrated circuits (ASIC), digital signal processor (DSP) or graphics processing unit (GPU) hardware arranged homogenously or heterogeneously. The memory **344** may be but is not limited to a random access memory (RAM), read only memory (ROM), or other electronic, optical, magnetic or any other computer readable medium.

The cloud database **340** also includes a communication device **346**. The communication device **346** may be capable of communication with the internet **306**. The communication device **346** may be configured to communicate with the cleaner computing device **500** through the internet **306**. The communication device **346** may be a software module that handles communications to and from the computer application **550** or to and from the space monitoring system **310**.

The cleaner computing device **500** may belong to or be in possession of a cleaning person **404**. The cleaning person **404** may be a janitor of the building **410**, a cleaner of the building **410**, a maintenance person of the building **410**, a building manager of the building **410**, or any other individual that may be responsible for the cleanliness within the building **410**.

The cleaner computing device **500** may be a desktop computer, a laptop computer, or a mobile computing device that is typically carried by a person, such as, for example a phone, a smart phone, a PDA, a smart watch, a tablet, a laptop, or any other mobile computing device known to one of skill in the art.

In an embodiment, the cleaner computing device **500** may be mobile device capable of augmented reality, such as, for example, a smart phone **500***a* or a pair of smart glasses **500***b*. The pair of smart glasses **500***b*, for example, may be designed in the shape of a pair of glasses with a lightweight wearable computer and a transparent display for handsfree work.

The cleaner computing device **500** includes a controller **510** configured to control operations of the cleaner computing device **500**. The controller **510** may be an electronic controller including a processor **530** and an associated memory **520** comprising computer-executable instructions (i.e., computer program product) that, when executed by the processor **530**, cause the processor **530** to perform various operations. The processor **530** may be, but is not limited to, a single-processor or multi-processor system of any of a wide array of possible architectures, including field programmable gate array (FPGA), central processing unit (CPU), application specific integrated circuits (ASIC), digital signal processor (DSP) or graphics processing unit (GPU) hardware arranged homogenously or heterogeneously. The memory **520** may be but is not limited to a random access memory (RAM), read only memory (ROM), or other electronic, optical, magnetic or any other computer readable medium.

It is understood that the computer application **550** may be a mobile application installed on the cleaner computing device **500**. The computer application **550** may be accessible from cleaner computing device **500**, such as, for example, a software-as-as service or a website. The computer application **550** may be in communication with the cloud database **340** via the internet **306**.

The cleaner computing device **500** includes a communication device **540** configured to communicate with the internet **306** through one or more wireless signals. The one or more wireless signals may include Wi-Fi, Bluetooth, Zigbee, Sub-GHz RF Channel, cellular, satellite, or any other wireless signal known to one of skill in the art. Alternatively, the cleaner computing device **500** may be connected to the internet **306** through a hardwired connection. The cleaner computing device **500** is configured to communicate with the cloud database **340** through the internet **306**.

The cleaner computing device **500** may include a display device **580**, such as for example a computer display, an LCD display, an LED display, an OLED display, a touchscreen of a smart phone, tablet, or any other similar display device known to one of the skill in the art. The cleaning person **404** operating the cleaner computing device **500** is able to view the computer application **550** through the display device **580**. If the cleaner computing device **500** is a pair of smart glasses **500***b*, then the display device **580** may be a transparent lens of the pair of smart glasses **500***b*.

The cleaner computing device **500** includes an input device **570** configured to receive a manual input from a user (e.g., human being) of cleaner computing device **500**. The input device **570** may be a keyboard, a touch screen, a joystick, a knob, a touchpad, one or more physical buttons, a microphone configured to receive a voice command, a camera or sensor configured to receive a gesture command, an inertial measurement unit configured to detect a shake of the cleaner computing device **500**, or any similar input device known to one of skill in the art. The user operating the cleaner computing device **500** is able to enter data into the computer application **550** through the input device **570**. The input device **570** allows the user operating the cleaner computing device **500** to data into the computer application **550** via a manual input to input device **570**. For example, the user may respond to a prompt on the display device **580** by entering a manual input via the input device **570**. In one example, the manual input may be a touch on the touchscreen. In an embodiment, the display device **580** and the input device **570** may be combined into a single device, such as, for example, a touchscreen on the smart phone **500***a*.

The cleaner computing device **500** device may also include a feedback device **560**. The feedback device **560** may activate in response to a manual input via the input device **570**. The feedback device **560** may be a haptic feedback vibration device and/or a speaker emitting a sound. The feedback device **560** may activate to confirm that the manual input entered via the input device **570** was received via the computer application **550**. For example, the feedback device **560** may activate by emitting an audible sound or vibrate the cleaner computing device **500** to confirm that the manual input entered via the input device **570** was received via the computer application **550**.

The cleaner computing device **500** may also include a location determination device **590** that may be configured to determine a location of the cleaner computing device **500** using cellular signal triangulation, a global position satellite (GPS), or any location termination method known to one of skill in the art.

The cleaner computing device **500** may also include an augmented reality capable camera **592** configured to allow the computer application **550** capture live images to perform augmented reality operations.

In an embodiment, the camera **320** may be able to determine a role of the individual **402** and differentiate cleaning person **404** in the interior space **412** to clean (e.g., a cleaning role) versus individuals **402** whose presence may require further cleaning (e.g., a contamination role). The instructions **386** detailing how to clean the object **420** that are transmitted to the cleaning person **404** may depend on how may individuals **418** with a contamination role were in a particular location and for how long.

The space monitoring system **310** is configured to utilize the space monitoring devices **320** to monitor a location of an individual **402** relative to an object **420** in the internal space **412** and duration of that individual **402** in the location. Then a determination may be made whether the object **420** requires a cleaning following the individual **402** being at that location for greater than a specific period of time. The space monitoring system **310** and/or the cloud database **340** may make the determination that the object **420** requires a cleaning. The determination on whether a cleaning is required may also be dependent on a number of individuals **402**.

The object **420** may be a table, desk, chair, seat, railing, handle, workstation, counter, doorknob, door handle, floor, carpet, rug, sink, light switch, touch surface, bar top, thermostat, computer, mobile device, conference room, cafeteria, coffee machine, vending machine, ATM, touch screen, office, or any other surface or area that may require cleaning after an individual being in that location for greater than a specific period of time.

Once the determination is made that an object **420** requires a cleaning then a cleaning request **380** may be transmitted to the cleaner computing device **500** so that it may be reviewed by the cleaning person **404** and then the cleaning person **404** may go clean the object **420**. The cleaning request **380** may include an object name **382**, a location **384** of the object **420**, and instructions **386** detailing how to clean the object **420**. The object name **382** may be a detailed description of the object **420**, such as, for example workstation, desk, table, or counter. The location **384** of the object **420** may be a specific location of an object, such as, for example, a latitude, longitude, and height where the object **420** is located. Alternatively, the location **384** of the object **420** may be a general or rough location of the object **420**, such as, for example, the object **420** is located on the third floor in room **414**. The instructions **386** may be detailed instructions on what to use to clean the object **420** and how long to clean the object **420**. The instructions **386** may vary based on how many individuals **402** were detected near the object **420** and for how long.

The cleaning person **404** may review the cleaning request **380** via the computer application **550** using the cleaner computing device **500**. The computer application **550** is configured to display the cleaning request **380** via the display device **580** and/or provide audio instructions for the cleaning request **380** via a feedback device **560** (e.g., speakers). The object name **382**, the location **384**, and the instructions **386** may all be displayed on the display device **580** or provided through audio instructions via the feedback device **560**.

The cleaner computing device **500** may convey the cleaning request **380** to the cleaning person **404** through augmented reality. For example, the cleaner computing device **500** may use the augmented reality capable camera **592** to capture live images of a space (e.g., the internal space **412**) and display those live images on the display device **580** while displaying the object name **382**, location **384**, and/or instructions **386**. The computer application **550** is configured to highlight or identify the objects **420** that require cleaning on the display device **580** using augmented reality while using the augmented reality capable camera **592**. Advantageously, the use of augmented reality will allow the objects **420** that require cleaning to be highlighted or identified on the display device **580** so that the cleaning person **404** can easily and quickly identified what objects **420** to clean. The object **420** that require cleaning may be highlight using text, symbols, colors, or any combination thereof.

The smart phone **500***a* may also be connected to a wearable camera **406** worn by the cleaning person **404** that captures images of how the cleaning person **404** is cleaning the object **420**, then the computer application **550** and/or the cloud database **340** can analyze the cleaning of the object **420** by the cleaning person **404** and provide further instructions as required. Alternatively, the augmented reality capable camera **592** may be configured to capture images of how the cleaning person **404** is cleaning the object **420**, then the computer application **550** and/or the cloud database **340** can analyze the cleaning of the object **420** by the cleaning person **404** and provide further cleaning instructions, compliments, or cleaning review as required. In one example, the further cleaning instructions may state to further disinfect the object **420** or a spot was missed. In another example, the compliments or cleaning review may state that "the object was cleaned appropriately, well done!!".

Alternatively, the augmented reality capable camera **592** may be configured to capture images of the object **420** after the cleaning person **404** has cleaned it, then the computer application **550** and/or the cloud database **340** transmit an image to an owner of the object to give proof to the owner that the object **420** was properly cleaned.

A computing device **600** may belong to or be in possession of the individual **402**. The individual **404** may be an employee of the building **410**, a resident of the building **410**, a visitor to the building **410**, or any other individual that may be within the building **410**.

The computing device **600** may be a desktop computer, a laptop computer, or a mobile computing device that is typically carried by a person, such as, for example a phone, a smart phone, a PDA, a smart watch, a tablet, a laptop, smart glasses, or any other mobile computing device known to one of skill in the art.

In an embodiment, the computing device **600** may be mobile device capable of augmented reality, such as, for example, a smart phone **600**. The computing device **600** may be the wearable tracking device **418**.

The computing device **600** includes a controller **610** configured to control operations of the computing device **600**. The controller **610** may be an electronic controller including a processor **630** and an associated memory **620** comprising computer-executable instructions (i.e., computer program product) that, when executed by the processor **630**, cause the processor **630** to perform various operations. The processor **630** may be, but is not limited to, a single-processor or multi-processor system of any of a wide array of possible architectures, including field programmable gate array (FPGA), central processing unit (CPU), application specific integrated circuits (ASIC), digital signal processor (DSP) or graphics processing unit (GPU) hardware arranged homogenously or heterogeneously. The memory **620** may be but is not limited to a random access memory (RAM), read only memory (ROM), or other electronic, optical, magnetic or any other computer readable medium.

It is understood that the computer application **650** may be a mobile application installed on the computing device **600**. The computer application **650** may be accessible from computing device **600**, such as, for example, a software-as-as service or a website. The computer application **650** may be in communication with the cloud database **340** via the internet **306**.

The computing device **600** includes a communication device **640** configured to communicate with the internet **306** through one or more wireless signals. The one or more wireless signals may include Wi-Fi, Bluetooth, Zigbee, Sub-GHz RF Channel, cellular, satellite, or any other wireless signal known to one of skill in the art. Alternatively, the computing device **600** may be connected to the internet **306** through a hardwired connection. The computing device **600** is configured to communicate with the cloud database **340** through the internet **306**.

The computing device **600** may include a display device **680**, such as for example a computer display, an LCD display, an LED display, an OLED display, a touchscreen of a smart phone, tablet, or any other similar display device known to one of the skill in the art. The cleaning person **404** operating the computing device **600** is able to view the computer application **650** through the display device **680**. If the computing device **600** is a pair of smart glasses **600***b*, then the display device **680** may be a transparent lens of the pair of smart glasses **600***b*.

The computing device **600** includes an input device **670** configured to receive a manual input from a user (e.g., human being) of computing device **600**. The input device **670** may be a keyboard, a touch screen, a joystick, a knob, a touchpad, one or more physical buttons, a microphone configured to receive a voice command, a camera or sensor configured to receive a gesture command, an inertial measurement unit configured to detect a shake of the computing device **600**, or any similar input device known to one of skill in the art. The user operating the computing device **600** is able to enter data into the computer application **650** through the input device **670**. The input device **670** allows the user operating the computing device **600** to data into the computer application **650** via a manual input to input device **670**. For example, the user may respond to a prompt on the display device **680** by entering a manual input via the input device **670**. In one example, the manual input may be a touch on the touchscreen. In an embodiment, the display device **680** and the input device **670** may be combined into a single device, such as, for example, a touchscreen on the smart phone **600***a*.

The computing device **600** device may also include a feedback device **660**. The feedback device **660** may activate in response to a manual input via the input device **670**. The feedback device **660** may be a haptic feedback vibration device and/or a speaker emitting a sound. The feedback device **660** may activate to confirm that the manual input entered via the input device **670** was received via the computer application **650**. For example, the feedback device **660** may activate by emitting an audible sound or vibrate the computing device **600** to confirm that the manual input entered via the input device **670** was received via the computer application **650**.

The computing device **600** may also include a location determination device **690** that may be configured to determine a location of the computing device **600** using cellular signal triangulation, a global position satellite (GPS), or any location termination method known to one of skill in the art.

The cloud database **340** may be configured to transmit an alert **389** to the computing device **600** when the cleaning request **380** is determined. The alert **389** may instruct the individual **402** to avoid the object **420** if cleaning is required. For example, the object **420** may be a door and the door may become contaminated if too many people interact with it, so the alert **389** may instruct the individual **418** to utilize another door that is not contaminated.

Referring now to FIG. **2**, with continued reference to FIG. **1**, a flow diagram illustrating an exemplary computer implemented method **800** for monitoring and controlling a cleanliness of a space (e.g., internal space **412**) is illustrated in accordance with an embodiment of the present disclosure. In embodiment, the method **800** is performed by the space cleanliness analysis system **300**.

At block **804**, a distance of one or more individuals **402** relative to an object **420** within the space is monitored using a space monitoring system **310**. The distance of one or more individuals **402** relative to the object **420** within the space may be monitored by detecting the distance of one or more individuals **402** relative to the object **420** within the space using a camera **322**, a door access control device **324**, and/or a wireless signal tracking device **326**. The wireless signal tracking device **326** may be configured to detect an advertisement of a wearable tracking device **418** worn by the one or more individuals **402** or wireless connect to the wearable tracking device **418** worn by the one or more individuals **402**.

At block **806**, it is determined when the distance is less than a selected distance. For example, the selected distance may be equal to zero meaning that the individual **402** is located at the object **420**.

At block **808**, a number of the one or more individuals **402** within the selected distance of object **420** is determined.

At block **810**, a duration that the one or more individuals **402** are within the selected distance of the object **420** is determined.

At block **812**, it may be determined that the object **420** requires a cleaning when at least one of: the number of the one or more individuals **402** within the selected distance of object **420** exceeds a threshold number, or the duration that the one or more individuals **402** are within the selected distance of the object **420** exceeds a threshold time. More individuals **402** may increase the likelihood that germs are spread on the object **420** and thus will require cleaning. The more time an individual **402** spends near the object **420** may increase the likelihood that germs are spread on the object **420** and thus will require cleaning.

The method **800** may include that a cleaning request **380** is generated. The method **800** may also include that the cleaning request **380** is transmitted to a computing device **500** of a cleaning person **404**. The cleaning request **380** may include at least one of an object name **382** of the object **420**, a location **384** of the object **420**, or instructions **386** detailing how to clean the object **420**. The cleaning request **380** may be provided to a cleaning person **404** through a computing device **500**. The cleaning request **380** may be displayed for a cleaning person **404** on a display device **580** of a computing device **500**.

The method **800** may further include an image of the object **420** is captured on an augmented reality capable camera **592** of the cleaner computing device **500** and then the image of the object **420** is displayed on a display device **580** of the cleaner computing device **500** while using the augmented reality capable camera **592**. It may then be identified using augmented reality the object **420** that requires the cleaning while using the augmented reality capable camera **592**. In an embodiment, the cleaner computing device **500** is a smart phone **500***a*. In another embodiment, the cleaner computing device **500** is a pair of smart glasses **500***b*.

The method **800** may further include that a cleaning of the object **420** performed by a cleaning person **404** is monitored using a wearable camera **406** or the augmented reality capable camera **592** of the pair of smart glasses **500***b*. Further cleaning instructions, a compliment, or a cleaning review based on the cleaning of the object **420** performed by the cleaning person **404** may be provided.

The method **800** may further include that an alert **389** is transmitted to a computing device **600** for an individual **402** to avoid the object **420** in response to determining that the object **420** requires the cleaning.

While the above description has described the flow process of FIG. **2** in a particular order, it should be appreciated that unless otherwise specifically required in the attached claims that the ordering of the steps may be varied.

As described above, embodiments can be in the form of processor-implemented processes and devices for practicing those processes, such as processor. Embodiments can also be in the form of computer program code (e.g., computer program product) containing instructions embodied in tangible media (e.g., non-transitory computer readable medium), such as floppy diskettes, CD ROMs, hard drives, or any other non-transitory computer readable medium, wherein, when the computer program code is loaded into and executed by a computer, the computer becomes a device for practicing the embodiments. Embodiments can also be in the form of computer program code, for example, whether stored in a storage medium, loaded into and/or executed by a computer, or transmitted over some transmission medium, such as over electrical wiring or cabling, through fiber optics, or via electromagnetic radiation, wherein, when the computer program code is loaded into and executed by a computer, the computer becomes a device for practicing the exemplary embodiments. When implemented on a general-purpose microprocessor, the computer program code segments configure the microprocessor to create specific logic circuits.

The term "about" is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, "about" can include a range of ±8% or 5%, or 2% of a given value.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

What is claimed is:

1. A method for monitoring and controlling a cleanliness of a space, the method comprising:

monitoring a distance of one or more individuals relative to an object within the space using a space monitoring system;

determining when the distance is less than a selected distance;

determining a number of the one or more individuals within the selected distance of object;

determining a duration that the one or more individuals are within the selected distance of the object; and

determining that the object requires a cleaning when at least one of:

the number of the one or more individuals within the selected distance of object exceeds a threshold number, or

the duration that the one or more individuals are within the selected distance of the object exceeds a threshold time;

generating a cleaning request;

providing the cleaning request to a cleaning person through a cleaner computing device;

capturing an image of the object on an augmented reality capable camera of the cleaner computing device;

displaying the image of the object on a display device of the cleaner computing device while using the augmented reality capable camera; and

identifying the object that requires the cleaning using augmented reality while using the augmented reality capable camera.

2. The method of claim **1**, further comprising:

transmitting the cleaning request to a computing device of a cleaning person.

3. The method of claim **2**, wherein the cleaning request comprises at least one of an object name of the object, a location of the object, or instructions detailing how to clean the object.

4. The method of claim **1**, wherein monitoring the distance of the one or more individuals relative to the object within the space using the space monitoring system further comprises:

detecting the distance of one or more individuals relative to the object within the space using a camera.

5. The method of claim **1**, wherein monitoring the distance of the one or more individuals relative to the object within the space using the space monitoring system further comprises:

detecting the distance of one or more individuals relative to the object within the space using a door access control device.

6. The method of claim **1**, wherein monitoring the distance of the one or more individuals relative to the object within the space using the space monitoring system further comprises:

detecting the distance of one or more individuals relative to the object within the space using a wireless signal tracking device.

7. The method of claim **6**, wherein the wireless signal tracking device is configured to detect an advertisement of a wearable tracking device worn by the one or more individuals or wireless connect to the wearable tracking device worn by the one or more individuals.

8. The method of claim **1**, further comprising:

displaying the cleaning request for a cleaning person on a display device of the cleaner computing device.

9. The method of claim **1**, wherein the cleaner computing device is a smart phone.

10. The method of claim **1**, wherein the cleaner computing device is a pair of smart glasses.

11. The method of claim **1**, further comprising:

monitoring the cleaning of the object performed by a cleaning person using a wearable camera.

12. The method of claim **11**, further comprising:

providing further cleaning instructions, a compliment, or a cleaning review based on the cleaning of the object performed by the cleaning person.

13. The method of claim **1**, further comprising:

monitoring a cleaning of the object performed by a cleaning person using an augmented reality capable camera of a pair of smart glasses.

14. The method of claim **13**, further comprising:

providing further cleaning instructions, a compliment, or a cleaning review based on the cleaning of the object performed by the cleaning person.

15. The method of claim **1**, further comprising:

transmitting an alert to a computing device for an individual to avoid the object in response to determining that the object requires the cleaning.

16. A space cleanliness analysis system for monitoring and controlling a cleanliness of a space, the space cleanliness analysis system comprising:

a processor; and

a memory comprising computer-executable instructions that, when executed by the processor, cause the processor to perform operations, the operations comprising:

monitoring a distance of one or more individuals relative to an object within the space using a space monitoring system;

determining when the distance is less than a selected distance;

determining a number of the one or more individuals within the selected distance of object;

determining a duration that the one or more individuals are within the selected distance of the object; and

determining that the object requires a cleaning when at least one of:

the number of the one or more individuals within the selected distance of object exceeds a threshold number, or

the duration that the one or more individuals are within the selected distance of the object exceeds a threshold time;

generating a cleaning request;

providing the cleaning request to a cleaning person through a cleaner computing device;

capturing an image of the object on an augmented reality capable camera of the cleaner computing device;

displaying the image of the object on a display device of the cleaner computing device while using the augmented reality capable camera; and

identifying the object that requires the cleaning using augmented reality while using the augmented reality capable camera.

17. A computer program product tangibly embodied on a non-transitory computer readable medium, the computer program product including instructions that, when executed by a processor, cause the processor to perform operations comprising:

monitoring a distance of one or more individuals relative to an object within the space using a space monitoring system;

determining when the distance is less than a selected distance;

determining a number of the one or more individuals within the selected distance of object;

determining a duration that the one or more individuals are within the selected distance of the object; and

determining that the object requires a cleaning when at least one of:

the number of the one or more individuals within the selected distance of object exceeds a threshold number, or

the duration that the one or more individuals are within the selected distance of the object exceeds a threshold time;

generating a cleaning request;

providing the cleaning request to a cleaning person through a cleaner computing device;

capturing an image of the object on an augmented reality capable camera of the cleaner computing device;

displaying the image of the object on a display device of the cleaner computing device while using the augmented reality capable camera; and

identifying the object that requires the cleaning using augmented reality while using the augmented reality capable camera.

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